

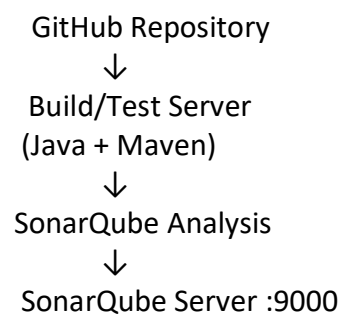
SonarQube Java Web Application Testing – Quick Guide

Objective

Set up SonarQube to analyze a Java Web Application using:

Server	Purpose
Build/Test Server	Build and test Java application
SonarQube Server	Host SonarQube on port 9000

Architecture



Step 1 – Create EC2 Servers

Create two Ubuntu 22.04 servers:

- Build/Test Server
- SonarQube Server

Recommended:

- t2.medium or higher
- Minimum 4 GB RAM

Step 2 – Configure Security Groups

Build/Test Server

- SSH – Port 22

SonarQube Server

- SSH – Port 22
- Custom TCP – Port 9000

Step 3 – Install Java, Maven, and Git on Build Server

```
sudo apt update
sudo apt install openjdk-17-jdk maven git -y
```

Verify:

```
java -version
mvn -version
git --version
```

Step 4 – Clone and Build Application

```
git clone <GITHUB-REPOSITORY-URL>
cd <PROJECT-FOLDER>
mvn clean install
```

Expected Output:

```
BUILD SUCCESS
```

Step 5 – Install Java on SonarQube Server

```
sudo apt update
sudo apt install openjdk-17-jdk unzip wget -y
```

Step 6 – Download and Install SonarQube

```
cd ~
wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-9.9.1.69595.zip
unzip sonarqube-9.9.1.69595.zip
mv sonarqube-9.9.1.69595 sonarqube
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@test-serv:~$ sudo wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.6.0.92116.zip
--2026-05-20 05:23:33-- https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.6.0.92116.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 3.170.51.90, 3.170.51.39, 3.170.51.3, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com)|3.170.51.90|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 731034761 (697M) [binary/octet-stream]
Saving to: 'sonarqube-10.6.0.92116.zip'

sonarqube-10.6.0.92116.zip  100%[=====] 697.17M  36.6MB/s   in 20s

2026-05-20 05:23:53 (35.2 MB/s) - 'sonarqube-10.6.0.92116.zip' saved [731034761/731034761]

ubuntu@test-serv:~$ ls
sonarqube-10.6.0.92116.zip
ubuntu@test-serv:~$ unzip
Command 'unzip' not found, but can be installed with:
sudo apt install unzip
ubuntu@test-serv:~$ sudo apt install unzip
Installing:
  unzip
Suggested packages:
  zip
```

```
creating: sonarqube-10.6.0.92116/elasticsearch/plugins/
ubuntu@test-serv:~$ ls
sonarqube-10.6.0.92116  sonarqube-10.6.0.92116.zip
ubuntu@test-serv:~$ sudo mv sonarqube-10.6.0.92116 sonarqube
ubuntu@test-serv:~$ ls
sonarqube  sonarqube-10.6.0.92116.zip
ubuntu@test-serv:~$ cd sonarqube/
ubuntu@test-serv:~/sonarqube$ ls
COPYING  bin  conf  data  dependency-license.json  elasticsearch  extensions  jres  lib  logs  temp  web
ubuntu@test-serv:~/sonarqube$ cd bin
ubuntu@test-serv:~/sonarqube/bin$ ls
elasticsearch  linux-x86-64  macosx-universal-64  windows-x86-64  winsw-license
ubuntu@test-serv:~/sonarqube/bin$ cd linux-x86-64/
ubuntu@test-serv:~/sonarqube/bin/linux-x86-64$ ls
sonar.sh
ubuntu@test-serv:~/sonarqube/bin/linux-x86-64$ ./sonar.sh
/usr/bin/java
Usage: ./sonar.sh { console | start | stop | force-stop | restart | status | dump }
ubuntu@test-serv:~/sonarqube/bin/linux-x86-64$ ls
sonar.sh
ubuntu@test-serv:~/sonarqube/bin/linux-x86-64$ ./sonar.sh start
/usr/bin/java
Starting SonarQube...
Started SonarQube.
ubuntu@test-serv:~/sonarqube/bin/linux-x86-64$ ./sonar.sh status
/usr/bin/java
SonarQube is running (4100).
```

Step 7 – Configure SonarQube

```
sudo sysctl -w vm.max_map_count=524288
```

Make permanent:

```
echo "vm.max_map_count=524288" | sudo tee -a /etc/sysctl.conf
```

Step 8 – Start SonarQube

```
cd ~/sonarqube/bin/linux-x86-64  
./sonar.sh start  
./sonar.sh status
```

Check Port 9000:

```
sudo ss -tulnp | grep 9000
```

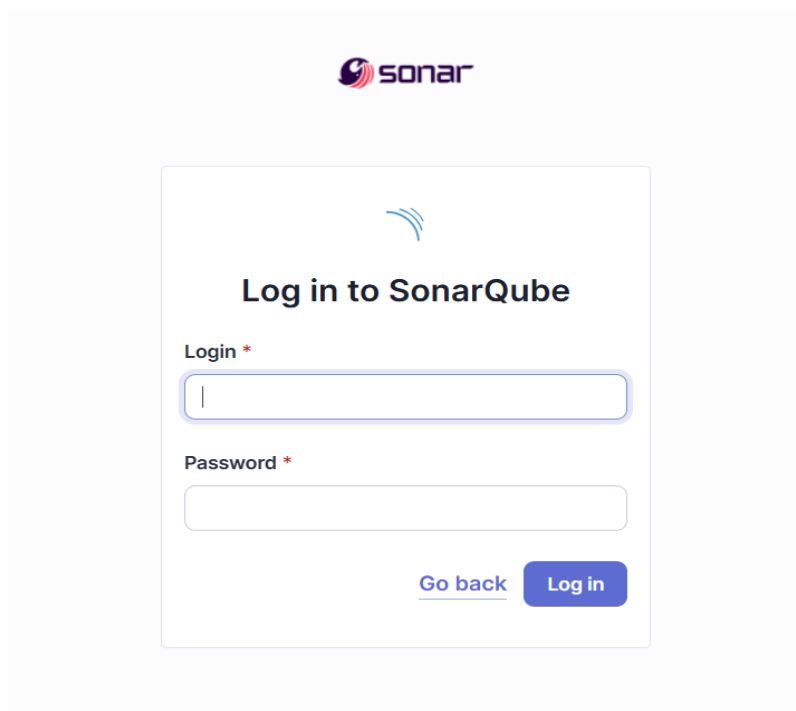
Step 9 – Access SonarQube

Open browser:

```
http://<SONARQUBE-SERVER-IP>:9000
```

Default Login:

Username	Password
admin	Admin

A screenshot of the SonarQube login page. At the top, there is the Sonar logo. Below it, a blue sonar icon is centered above the text "Log in to SonarQube". There are two input fields: "Login *" and "Password *". Below the "Login" field is a "Go back" link and a "Log in" button.

Generate a token from:

Tokens of Administrator

Generate Tokens

Name

Expires in

30 days

Generate

Name	Type	Project	Last use	Created	Expiration
------	------	---------	----------	---------	------------

kiran

User

3 hours ago

May 20, 2026

June 19, 2026

Revoke

Close

Step 10 – Install SonarScanner on Build Server

```
cd /opt
sudo wget https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-6.0.0.4432-linux.zip
sudo unzip sonar-scanner-cli-6.0.0.4432-linux.zip
sudo mv sonar-scanner-6.0.0.4432-linux sonar-scanner
```

Configure PATH:

```
sudo vi /etc/profile
```

Add:

```
export SONAR_SCANNER_HOME=/opt/sonar-scanner
export PATH=$PATH:$SONAR_SCANNER_HOME/bin
```

Reload:

```
source /etc/profile
```

Verify: sonar-scanner -v

Step 11 – Run SonarQube Analysis

```
cd ~/JavaWebCalculator
```

Run:

```
mvn sonar:sonar \
-Dsonar.projectKey=java-web-app \
-Dsonar.host.url=http://<SONARQUBE-SERVER-IP>:9000 \
-Dsonar.login=<SONAR-TOKEN>
```

Expected Output:

BUILD SUCCESS



```
Last login: Wed May 20 06:08:40 2026 from 104.28.155.21
ubuntu@build:~$ ls
JavaWebCalculator
ubuntu@build:~$ cd JavaWebCalculator/
ubuntu@build:~/JavaWebCalculator$ ls
pom.xml  src
ubuntu@build:~/JavaWebCalculator$ mvn clean verify sonar:sonar \
-Dsonar.projectKey=my-java-app \
-Dsonar.host.url=http://3.145.61.145:9000 \
-Dsonar.login=squ_4214c47c13b6ee3a48f1e07945967b571dff0168

[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 36 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 25 k
b/s)
```

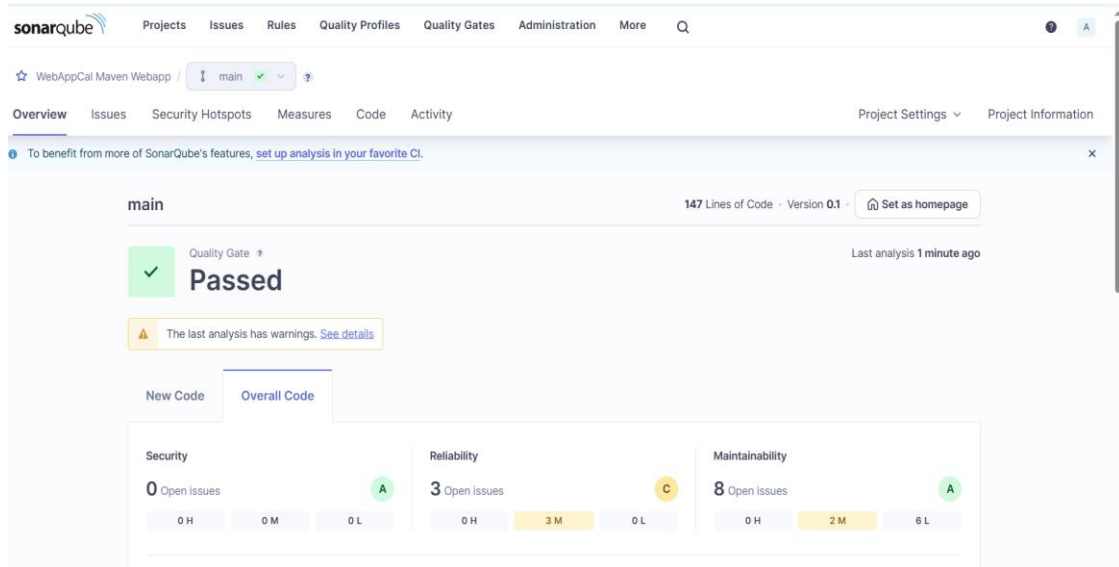
Step 12 – Verify Dashboard

Open:

<http://<SONARQUBE-SERVER-IP>:9000>

You can now view:

- Bugs
- Vulnerabilities
- Code Smells
- Coverage
- Quality Gate Status



Conclusion

This setup enables:

- Java application build and testing
- SonarQube code quality analysis
- Centralized dashboard for bugs, vulnerabilities, and code quality metrics